

AIM:

To write an assembly language program to implement 16 bit division using 8085 processor.

ALGORITHM:

- 1) Read dividend (16 bit).
- 2) Read divisor.
- 3) Count $\leftarrow 8$.
- 4) Left shift dividend.
- 5) Subtract divisor from upper-8 bits of dividend.
- 6) If $CS = 1$ go to 9.
- 7) Restore dividend.
- 8) Increment lower 8-bits of dividend.
- 9) Count $\leftarrow$ Count - 1.
- 10) If Count = 0 go to 5.
- 11) Store upper 8-bit dividend as remainder and lower 8-bit as quotient.
- 12) Stop.

PROGRAM:

MNEMONICS	EXPLANATION
LDA 8501	loads Accumulator with contents of 8501-H

MOV B, A

moves contents of A to B.

LDA 8500

loads A with 8500H

MVI C, 00

loads the immediate value of 00H to C.

LOOP: CMP B

the beginning of a loop, compares contents of register A with register B.

JC LOOP1

jumps to the LOOP1 label if the carry flag is set (i.e., if A is less than B after the comparison).

SUB B

subtracts the contents of B from A & stores in A.

INR C

increments register C by 1.

JMP LOOP

jump back to LOOP label, effectively continuing loop.

STA 8503

stores contents of A into 8503H.

DCR C

decrements contents of C by 1.

MOV A, C

moves C to A.

LOOP1:

Marks the continuation point of program after jump, stores the contents of A to 8502H.

RST 1

a software interrupt & jump to 0024H.

INPUT :

ADDRESS	DATA
8051	20
8050	2

OUTPUT:

ADDRESS	DATA
8502	10
8503	2

RESULT : 

Thus, the program was executed successfully using 8085 simulator.



Registers

A 00
BC 00 00
DE 00 00
HL 00 00
PSW 00 00
PC 42 22
SP 00 0A
Int-Reg 00

Flag

S 0
Z 1
AC 0
P 1
C 0

Load me at

```
1  LHLD 2050
2  SPHL
3  LHLD 2052
4  XCHG
5  LXI H,0000H
6  LXI B,0000H
7  AGAIN: DAD SP
8  JNC START
9  INX B
10 START: DCX D
11 MOV A,E
12 ORA D
13 JNZ AGAIN
14 SHLD 2054
15 MOV L,C
16 MOV H,B
17 SHLD 2055
18 HLT
```

Decimal - Hex Conversion

Decimal Hex

0 0
→ To Hex ← To Dec

I/O Ports

0 - + 00
Update Port Value

Memory

2052 - + 05
Update Memory

Data Stack KeyPad Memory I/O Ports

Start 2054 OK

Address (Hex)	Address	Data
0806	2054	50
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0
0810	2064	0
0811	2065	0
0812	2066	0
0813	2067	0
0814	2068	0
0815	2069	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

